Functional Linear Models with Functional Responses and Covariates

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Overview

- The fundamental question: If we have a covariate $z(s)$, how much of its variation over s should we use to fit the response $y(t)$ at fixed t?
- In the simplest case, we only use $z(t)$. We call this the **concurrent model** because it predicts y at time t by z at time t.
- If we want to use the behavior of z over an interval of values s, or over all values, things are more complex because this model has, effectively, an infinite amount of fitting power.

The full model for log precipitation

- We now want to predict the log precipitation profile $\text{LogPrec}_i(t)$ at time t from the entire temperature profile $\text{Temp}_i(s)$.
- The full model is

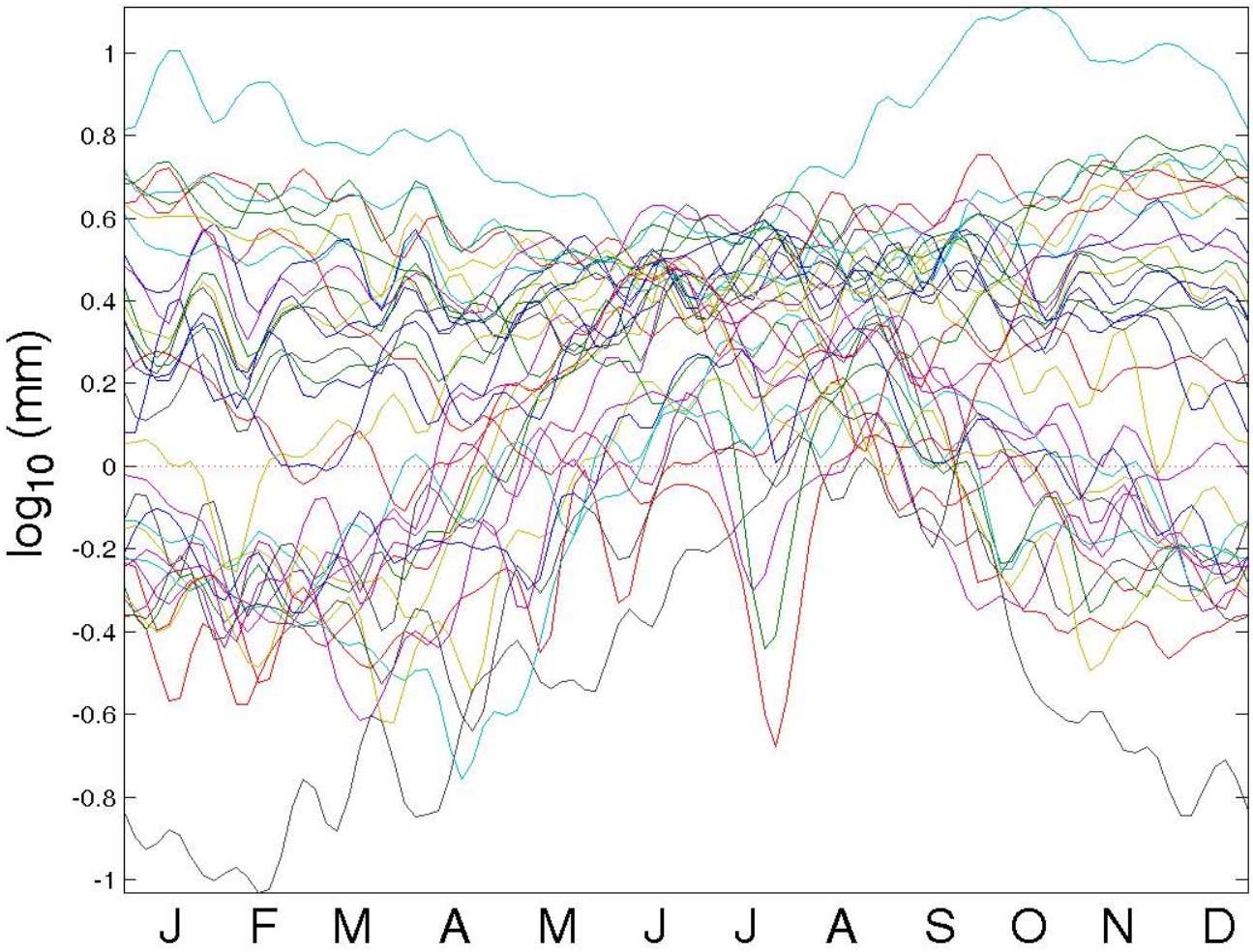
$$\text{LogPrec}_i(t) = \alpha(t) + \int_0^{365} \text{Temp}_i(s)\beta(s,t)ds + \epsilon_i(t).$$

- $\beta(s,t)$ indicates the influence of temperature at time s on precipitation at time t.

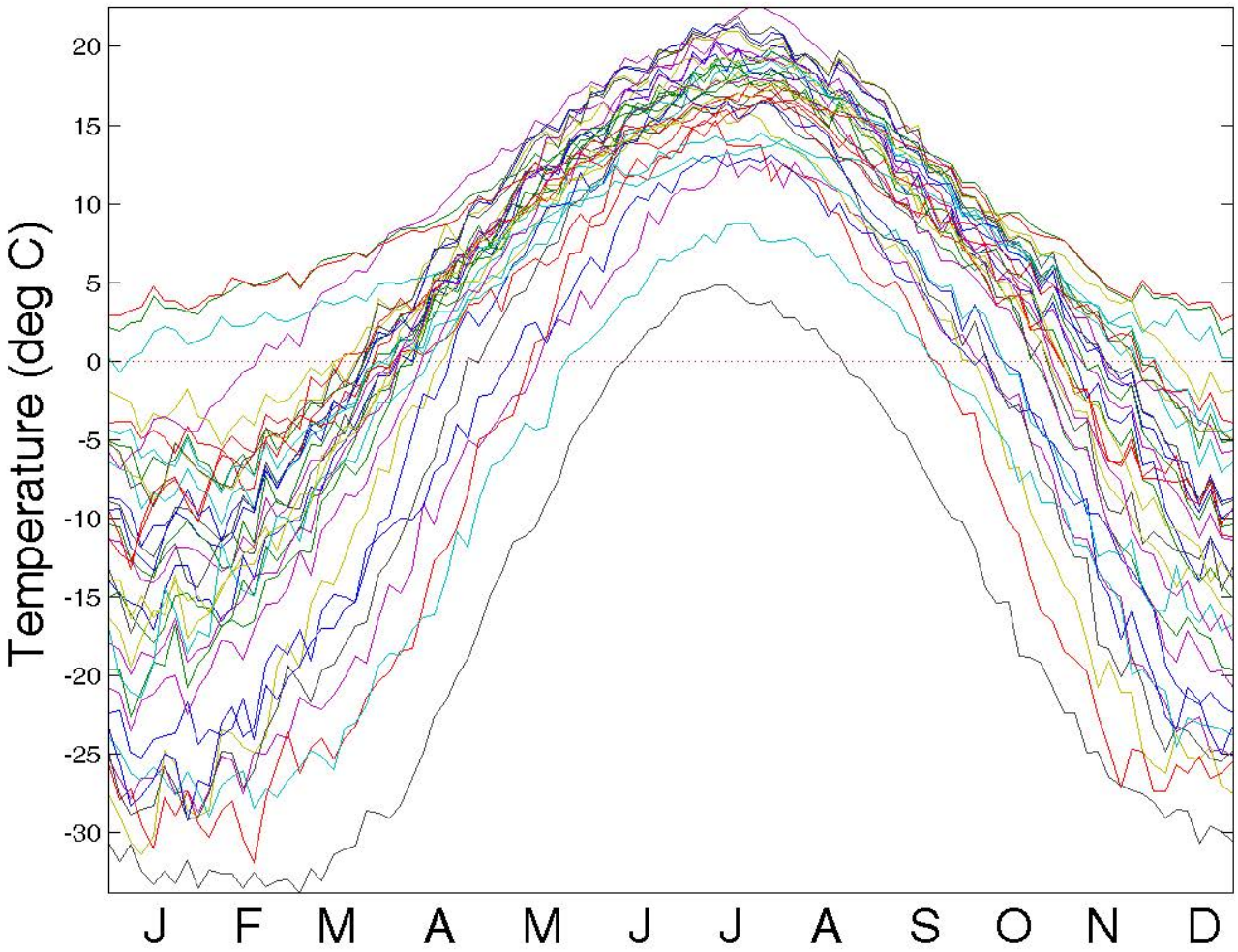
Reasons for using the whole temperature profile

- Temperatures at $s < t$ are surely informative for precipitation at time t.
- Influence of temperature over precipitation is complicated across different climate zones:
  - continental stations, where winters are cold and summers are hot, have most of their precipitation in the summer months.
  - Atlantic stations tend to have a fairly even distribution of precipitation over the whole year.
- Temperatures at $s > t$ also included since the weather patterns are periodic and we can predict rainfall in January by using temperature information in December.

Log precipitation functions



Temperature functions



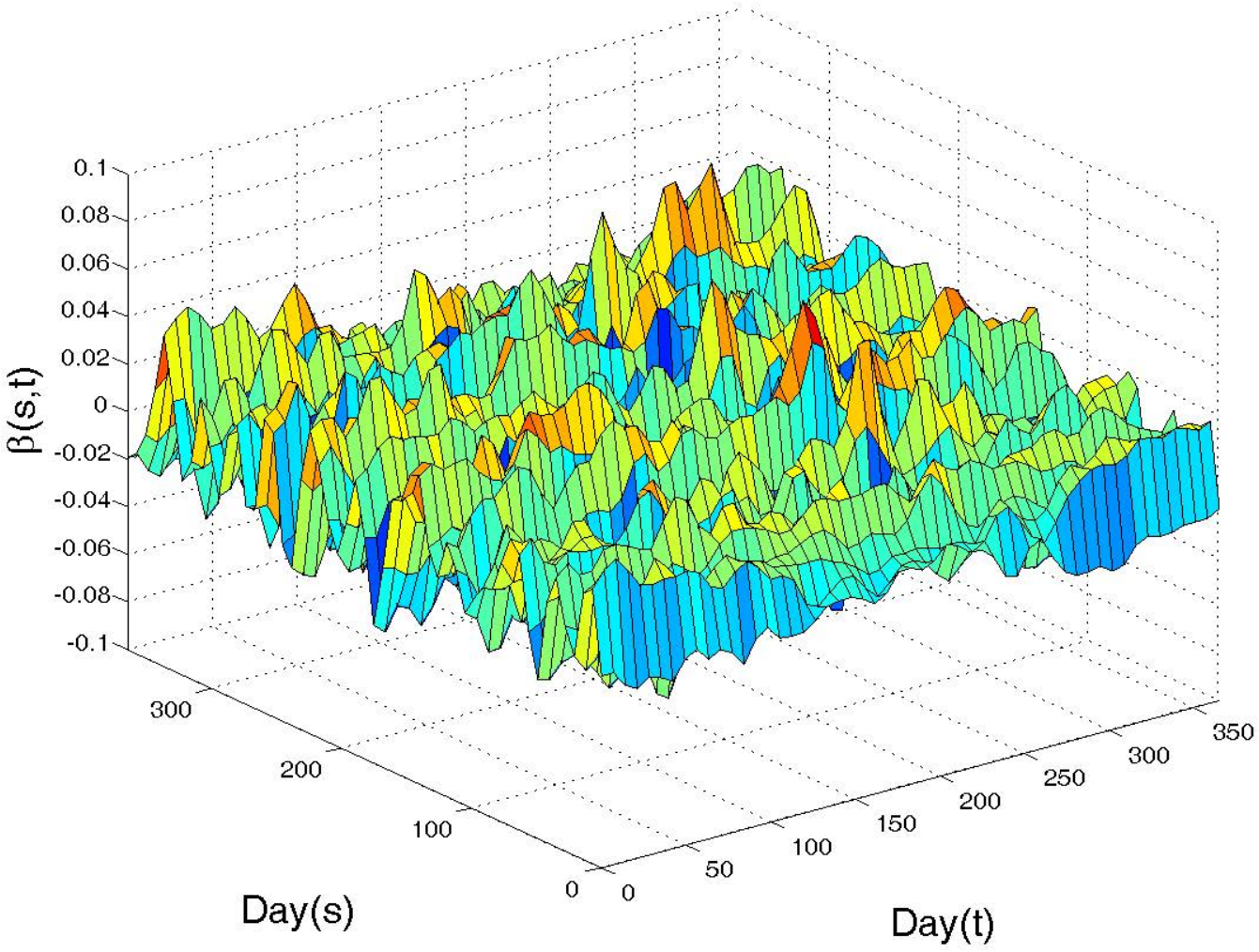
Roughness penalty on $\beta(s, t)$

- We have already learned from predicting total log precipitation that we will have to apply a roughness penalty to $\beta(s, t)$ as a function of s.
- What about its variation as a function of t?

Roughness penalty on $\beta(s, t)$

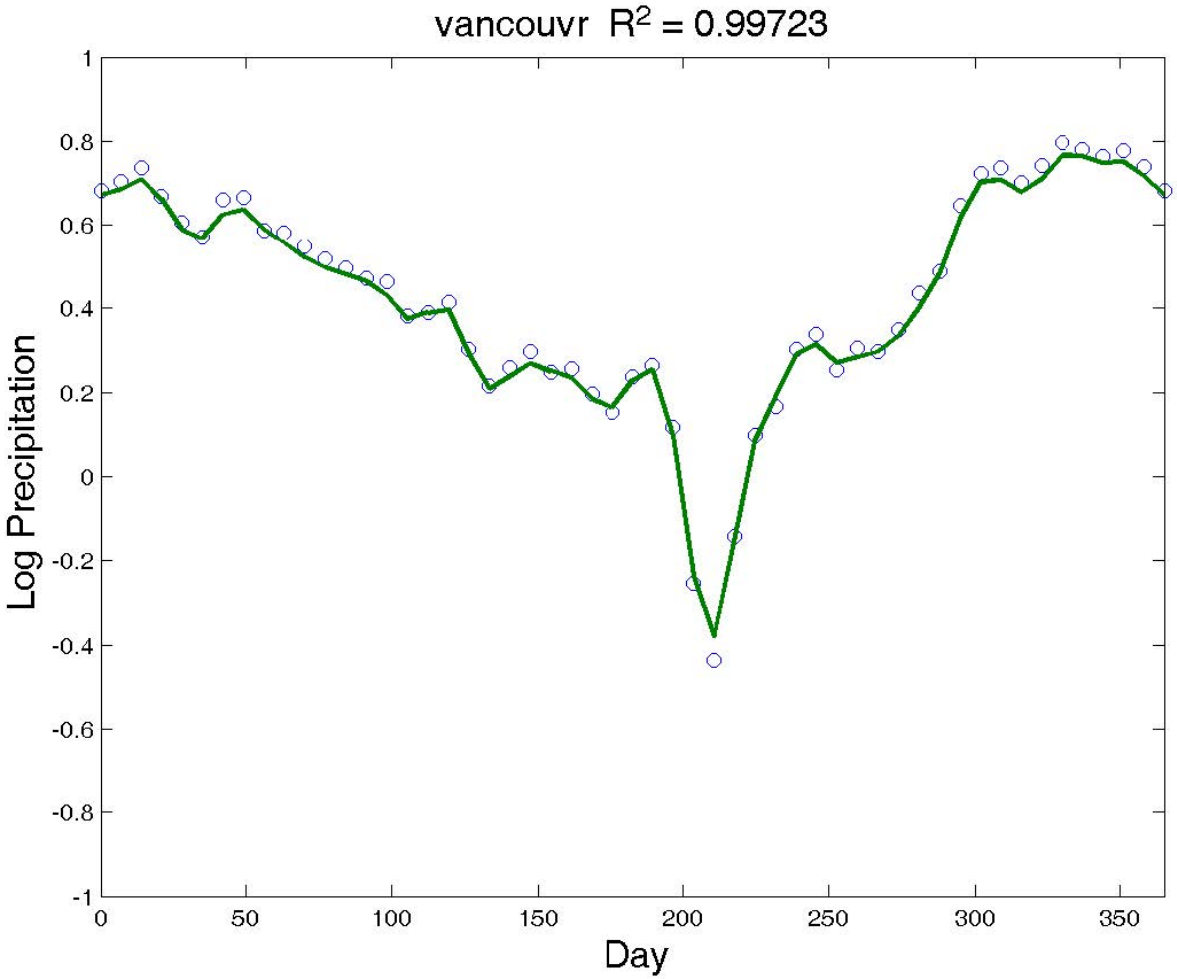
- We apply two harmonic acceleration roughness penalties to $\beta(s, t)$, one for its variation in s, and one for its variation in t.
- We'll look at $\beta(s, t)$ and at the fit to the log precipitation data for Vancouver.
- Let's see what happens with fairly light penalties on both types of variation.

Light penalties on s and t: $\hat{\beta}(s,t)$



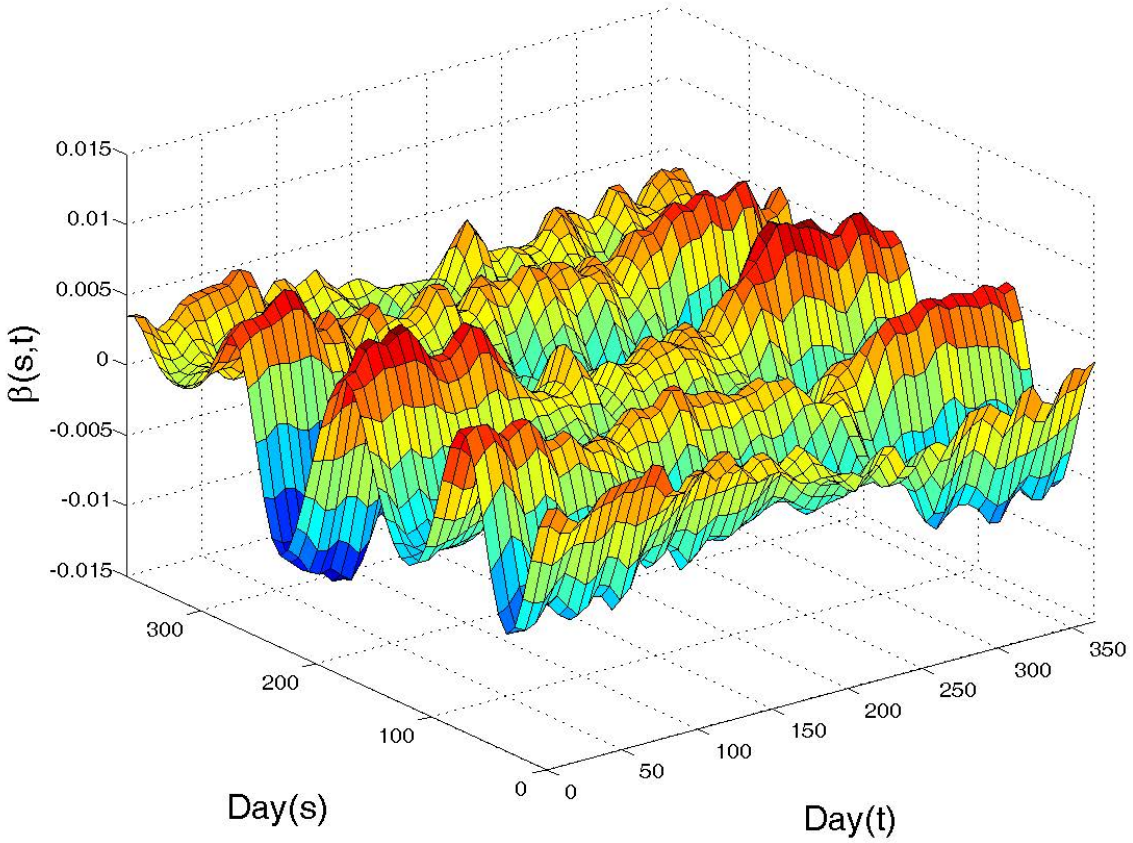
- $\beta(s,t)$ is impossible to interpret.

Light penalties on s and t: Vancouver



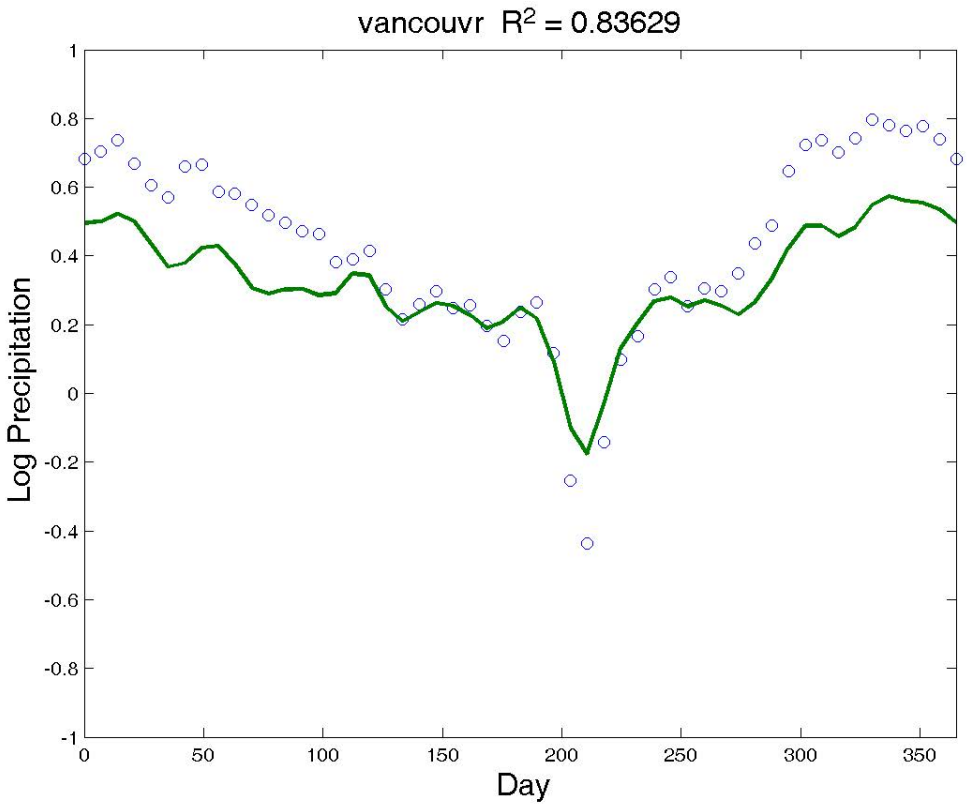
- we seem to have overfitted Vancouver's data.
- Let's increase the smoothing parameter for s.

Heavy penalty on s and light on t: $\hat{\beta}(s,t)$



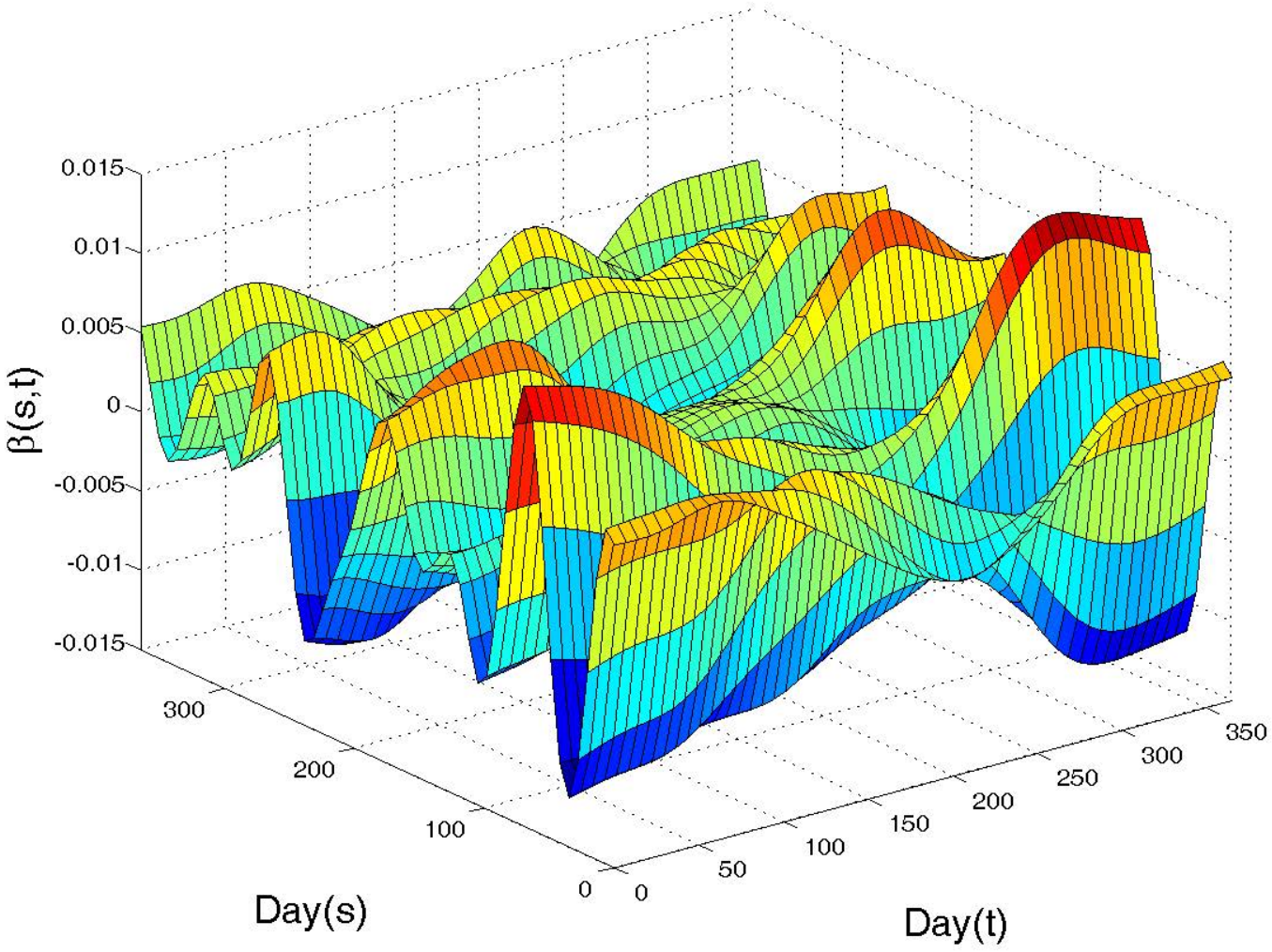
- $\beta(s,t)$ is interpretable as a function of s but impossible to understand in t.

Heavy penalty on s and light on t: Vancouver



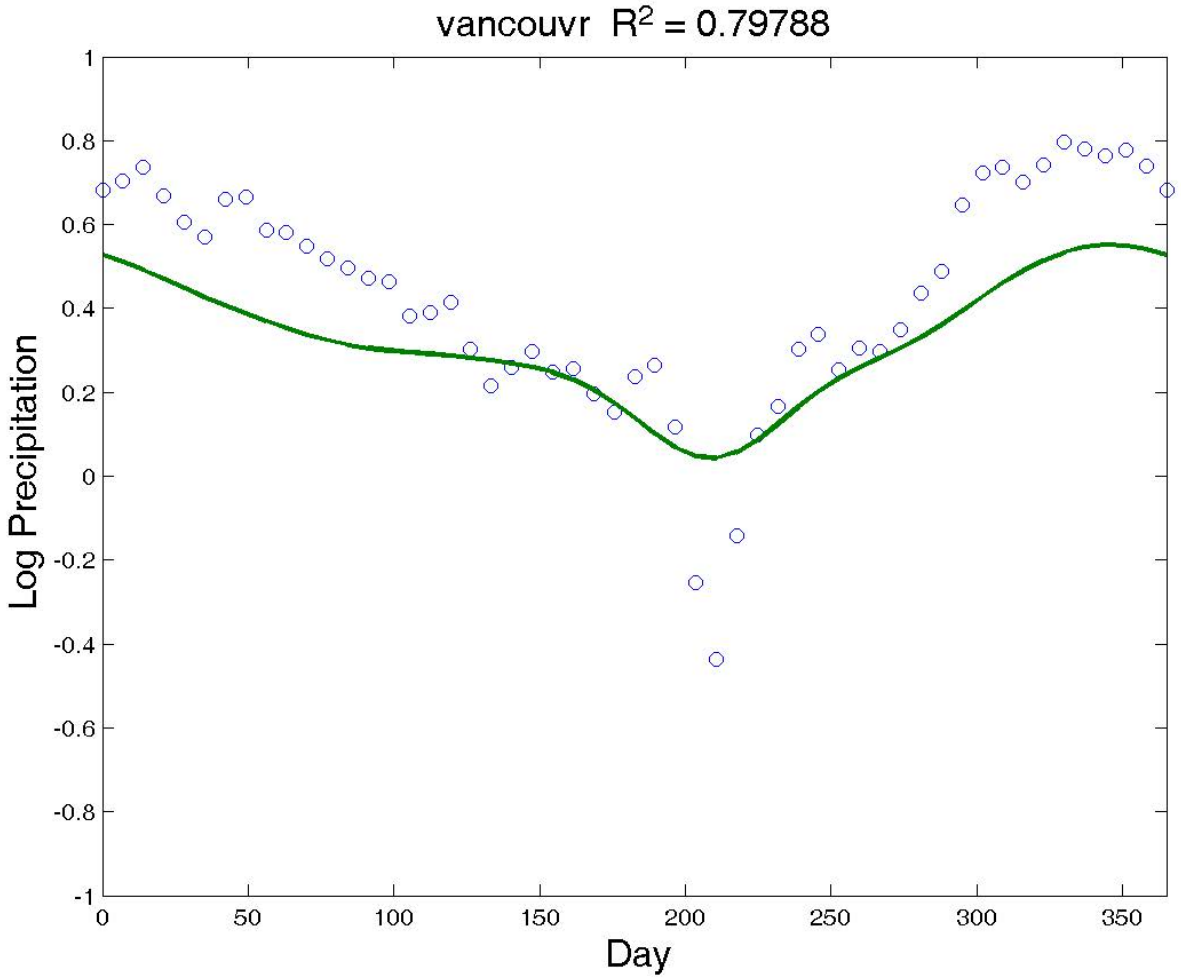
- We now have a more reasonable fit to Vancouver's data, but the fitting function is too rough.
- Let's increase smoothing parameters for both s and t.

Heavy penalties on s and t: $\hat{\beta}(s,t)$



- $\beta(s,t)$ is now smooth in both s and t.

Heavy penalties on s and t: Vancouver

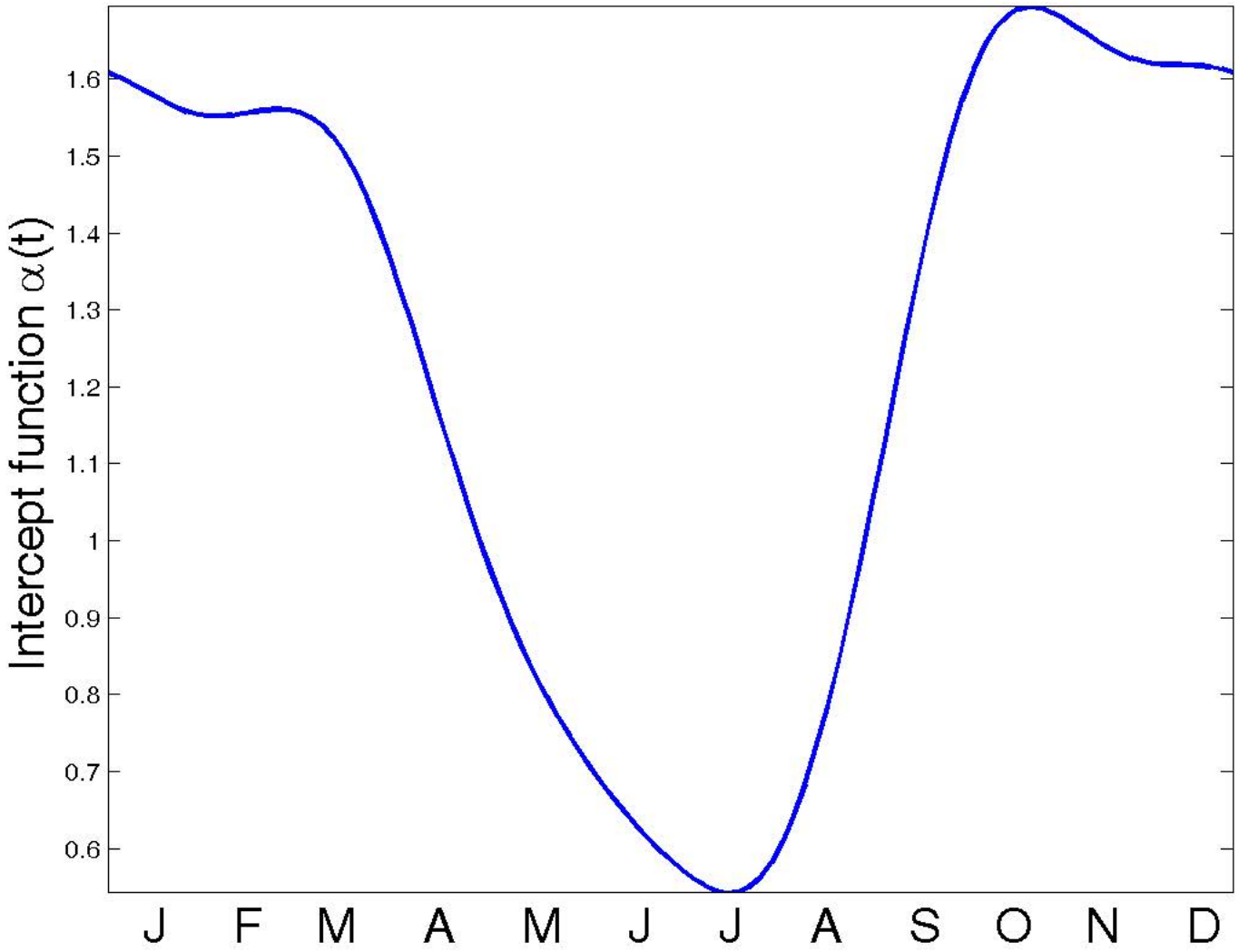


- The fit is reasonable and also smooth.

What do we see?

- Penalizing the roughness of $\beta(s,t)$ as a function of s prevents over-fitting.
- Penalizing the roughness of $\beta(s,t)$ as a function of t allows us to see how the influence of temperature on precipitation varies from one time to another.
- We can now see that temperature is much more influential in the winter than in the summer.
- The rapid oscillation in s suggests that it is a **derivative** of temperature that really influences precipitation.

The intercept function $\alpha(t)$



Remove the intercept function

Consider drop the intercept function $\alpha(t)$.

- From the normal equations,

$$\alpha(t) = \bar{y}(t) - \int \bar{z}(s)\beta(s,t)ds.$$

- Define $y_i^*(t) = y_i(t) - \bar{y}(t)$ and $z_i^*(s) = z_i(s) - \bar{z}(s)$.
- The full model can be rewritten as

$$y_i^*(t) = \int z_i^*(s)\beta(s,t)ds + \epsilon_i(t).$$

Basis expansion

- Expand $\beta(s,t)$ as:

$$\beta(s,t) = \boldsymbol{\theta}'(s)\mathbf{B}\boldsymbol{\eta}(t),$$

where the basis system $\boldsymbol{\theta}$ has K_1 functions and $\boldsymbol{\eta}$ has K_2 functions, and $\mathbf{B}$ is a K_1 by K_2 coefficient matrix.

- Define the N by K_1 matrix

$$\mathbf{Z}^* = \int \mathbf{z}^*(s)\boldsymbol{\theta}'(s)ds.$$

- The model in vector-matrix form:

$$\mathbf{y}^*(t) = \mathbf{Z}^*\mathbf{B}\boldsymbol{\eta}(t) + \boldsymbol{\epsilon}(t).$$

Roughness penalty

- Define the matrices $\mathbf{J}_{\eta\eta} = \int \boldsymbol{\eta}(t)\boldsymbol{\eta}'(t)dt$, $\mathbf{J}_{\theta\theta} = \int \boldsymbol{\theta}(t)\boldsymbol{\theta}'(t)dt$ and $\mathbf{J}_{\theta\eta} = \int \boldsymbol{\theta}(t)\boldsymbol{\eta}'(t)dt$.
- Penalty on s:

$$\begin{aligned}
\mathrm{PEN}_s(\beta) &= \int\int [L_s(\beta(s,t))]^2 dsdt \\
&= \int\int [L_s\boldsymbol{\theta}'(s)]\mathbf{B}\boldsymbol{\eta}(t)\boldsymbol{\eta}'(t)\mathbf{B}'[L_s\boldsymbol{\theta}(s)]dsdt \\
&= \int \mathrm{trace}[\mathbf{B}\boldsymbol{\eta}(t)\boldsymbol{\eta}'(t)\mathbf{B}'\mathbf{R}]dt. \\
&= \mathrm{trace}[\mathbf{B}'\mathbf{R}\mathbf{B}\mathbf{J}_{\eta\eta}],
\end{aligned}$$

  where $\mathbf{R} = \int [L_s\boldsymbol{\theta}(s)][L_s\boldsymbol{\theta}'(s)]ds$.
- Similarly, the penalty on t is: $\mathrm{PEN}_t(\beta) = \mathrm{trace}[\mathbf{B}'\mathbf{J}_{\theta\theta}]\mathbf{S}\mathbf{B}$, where $\mathbf{S} = \int [L_t\boldsymbol{\eta}(t)][L_t\boldsymbol{\eta}'(t)]dt$.

The estimate of $\mathbf{B}$

- Adding the two penalties, each multiplied by their respective smoothing parameters, the equation for $\mathbf{B}$ becomes

$$\mathbf{Z}^{*'}\mathbf{Z}^{*}\mathbf{B}\mathbf{J}_{\eta\eta} + \lambda_s \mathbf{R}\mathbf{B}\mathbf{J}_{\eta\eta} + \lambda_t \mathbf{J}_{\theta\theta}\mathbf{B}\mathbf{S} = \mathbf{Z}^{*'}\int \mathbf{y}\boldsymbol{\eta}'.$$

- This is equivalent to

$$[\mathbf{J}_{\eta\eta} \otimes (\mathbf{Z}^{*'}\mathbf{Z}^{*}) + \lambda_s \mathbf{J}_{\eta\eta} \otimes \mathbf{R} + \lambda_t \mathbf{S} \otimes \mathbf{J}_{\theta\theta}]\mathrm{vec}(\mathbf{B}) = \mathrm{vec}(\mathbf{Z}^{*'}\int \mathbf{y}\boldsymbol{\eta}').$$

Confidence intervals

- To compute standard errors, expand

$$\mathbf{y} = \mathbf{C}\phi,$$

where ϕ is a system of K_y basis functions and $\mathbf{C}$ is the N by K_y coefficient matrix.

- The estimate for $\mathbf{B}$ is:

$$\text{vec}(\hat{\mathbf{B}}) = [\mathbf{J}_{\eta\eta} \otimes (\mathbf{Z}^{*'}\mathbf{Z}^{*}) + \lambda_s \mathbf{J}_{\eta\eta} \otimes \mathbf{R} + \lambda_t \mathbf{S} \otimes \mathbf{J}_{\theta\theta}]^{-1} (\mathbf{J}_{\theta\eta} \otimes \mathbf{Z}^{*'})\text{vec}(\mathbf{C}).$$

- Hence the mapping from $\mathbf{C}$ to $\mathbf{B}$ is

$$[\mathbf{J}_{\eta\eta} \otimes (\mathbf{Z}^{*'}\mathbf{Z}^{*}) + \lambda_s \mathbf{J}_{\eta\eta} \otimes \mathbf{R} + \lambda_t \mathbf{S} \otimes \mathbf{J}_{\theta\theta}]^{-1} (\mathbf{J}_{\theta\eta} \otimes \mathbf{Z}^{*'}).$$

The historical model

- We were able to use all of $z(s)$ to predict $y(t)$ in the weather example because the data were periodic.
- In non-periodic situations, it would only be meaningful to use the values of s **up to** t.

$$y(t) = \alpha(t) + \int_{t-\delta(t)}^{t} z(s)\beta(s,t)ds + \epsilon(t).$$

- For the lower limit of integration $t - \delta(t)$, the width $\delta(t)$ of the interval of integration can vary over t or can be constant.
- We call this the *historical linear model*.

A general model

- More generally, we can integrate over a set Ω_t that varies with t,
- and also permit the covariate function z to vary over t as well,

$$y(t) = \alpha(t) + \int_{\Omega_t} z(s,t)\beta(s,t)ds + \epsilon(t).$$

The support domain of a historical model

- The support domain of a historical model is a subset of the triangular $\{(s,t) : 0 \leq s \leq t\}$.
- How do we construct basis function systems for such complex regions of integration?
- The triangular mesh algorithms used in *finite element methods* to solve partial differential equations are natural here.
- Triangular meshes adapt well to nonstandard boundaries.
- Triangular basis functions also can be regularized.
- The PDEtools toolbox in Matlab contains powerful functions for mesh generation.

The reference and background

- Reference: Malfait and Ramsay (2003). "The historical functional linear model". *Canadian Journal of Statistics* **31**(2): 115-128.
- Speech production can involve over 100 muscles from different parts of the body.
- The timing of the activation of different groups of muscles is a central issue for the anatomy and physiology of speech.
- Electromyograph (EMG) provides a noninvasive way to collecting data on muscle activation.

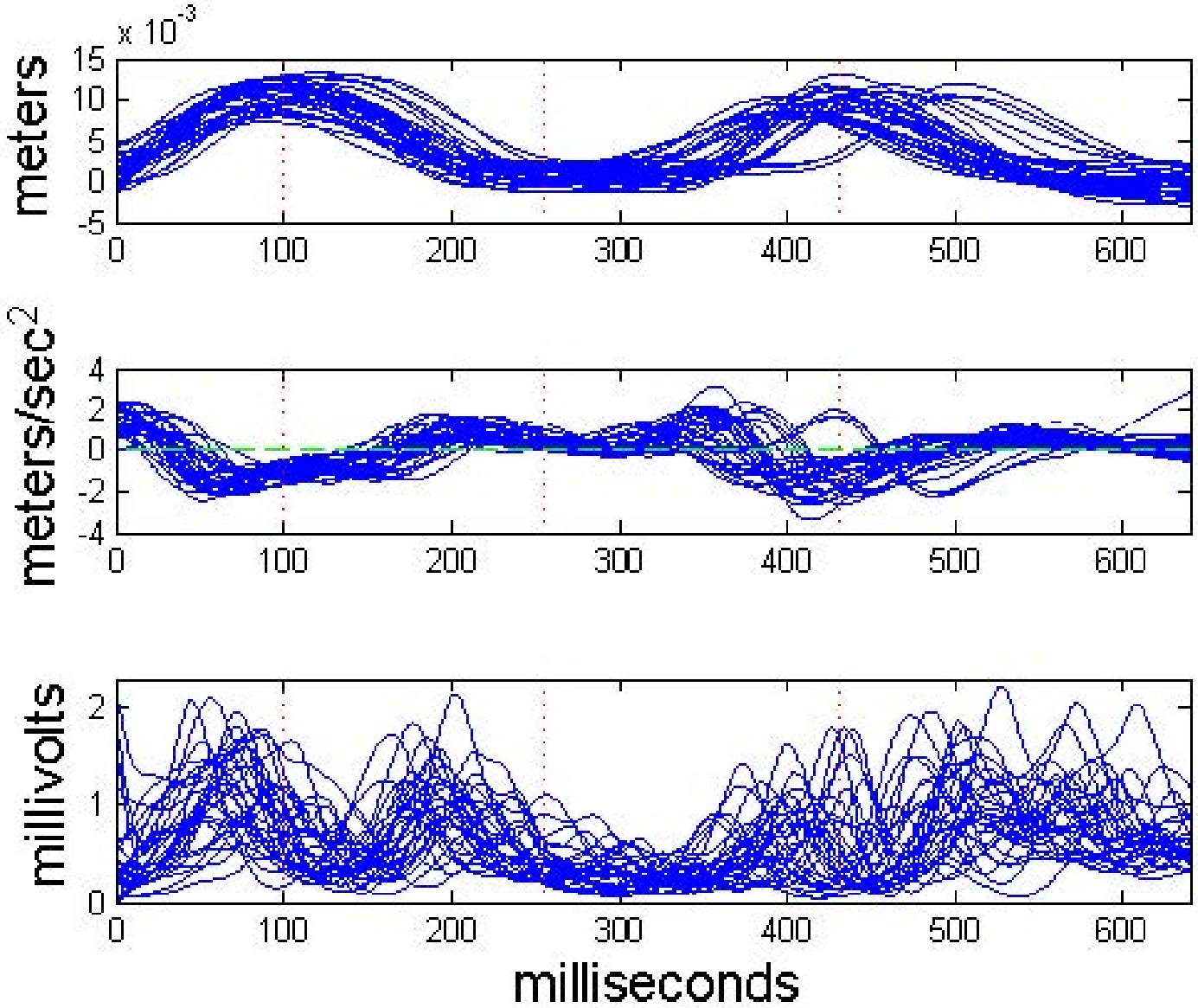
How does the EMG work here?

- A resting muscle generates no current.
- When a muscle is stimulated, the resulting activation travels along the muscle as waves of action potentials.
- With electrodes attached to the skin over the muscle, EMG can pick up these potentials.
- A complication: the time taken for a neural signal to be transduced into muscle contraction can vary considerably.
- For the facial muscles, a representative value is about 50 msec.

The data in the paper

- A subject was required to say the syllable "bob" $N = 32$ times.
- The duration of the syllable varied but was time-normalized to 700 msec.
- The plots shown in the next slides:
  - Top panel: the movement of the center of the lower lip.
  - Middle panel: the accelerations of the center of the lower lip, denoted by $y_i(t)$.
  - Lower panel: the so-called linear envelopes of EMG activities, denoted by $x_i(s)$, recorded in the primary muscle depressing the lower lip, the depressor labii inferior (DLI).

Data on lip movements



More background

- The lower lip trajectory can be segmented into roughly five epochs (separated by the dotted lines in the plots).
- The central interval corresponds to the /o/, during which the lip is stationary.
- To produce each /b/, the lip moves up and down.
- The DLI muscle plays two roles:
  - antagonist: when it brakes the movement during the ascending phrases; reflected by EMG bursts as the acceleration crosses zero while moving from a positive to a negative phase.
  - agonist: when it accelerates the lip during the descending phrases; reflected by EMG bursts at the start of the lip's descents.

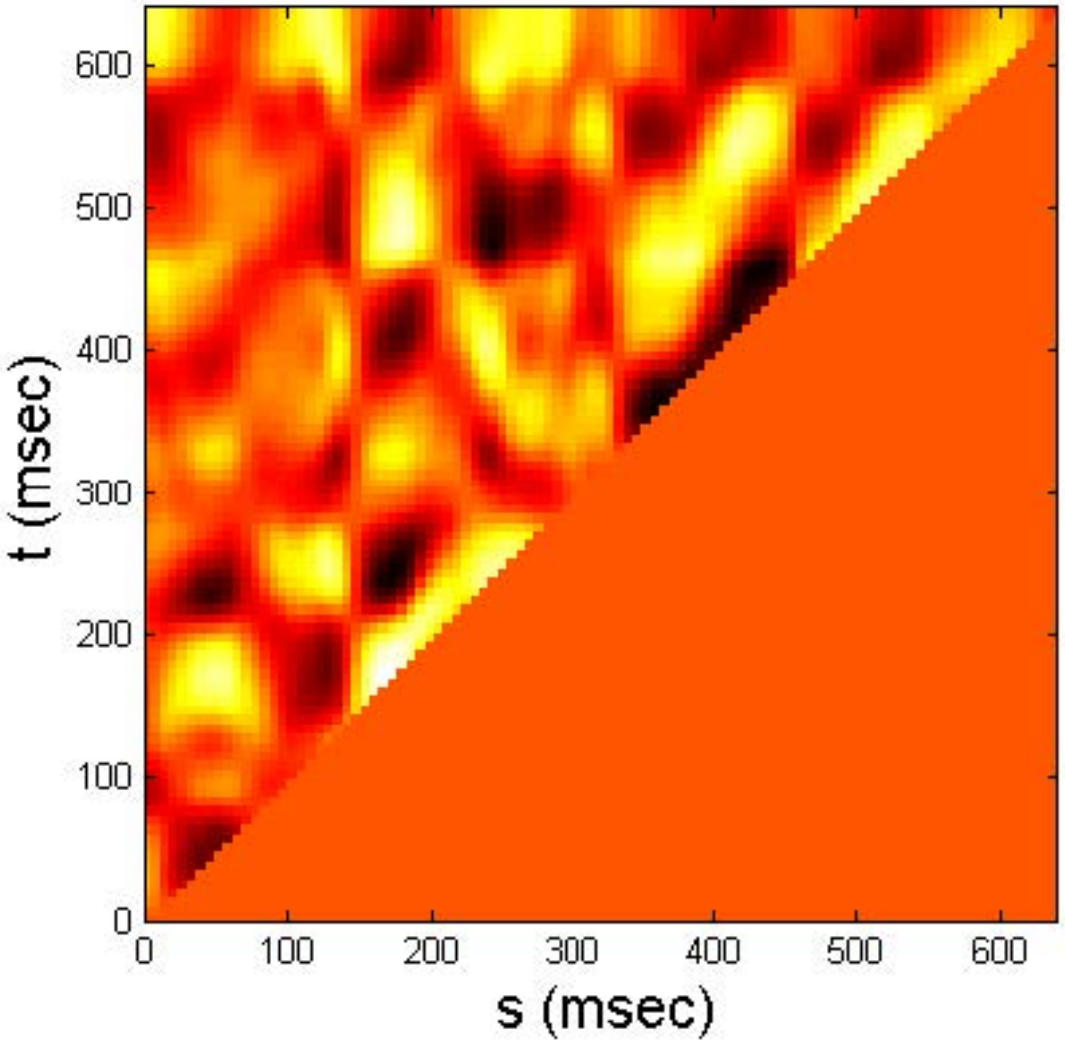
The feed-forward model

- Assume only feed-forward effects of EMG on lip acceleration, the model is

$$\mathrm{Lip}(t) = \alpha(t) + \int_{t-\delta(t)}^{t} \mathrm{EMG}(s)\beta(s,t)ds + \epsilon(t).$$

- Regression function $\beta(s,t)$ is defined over the triangular region $\{(s,t) : 0 \leq s \leq t \leq 0.7\}$.
- How far back should we allow for an influence? How large should the lag δ be?

The autocorrelation surface

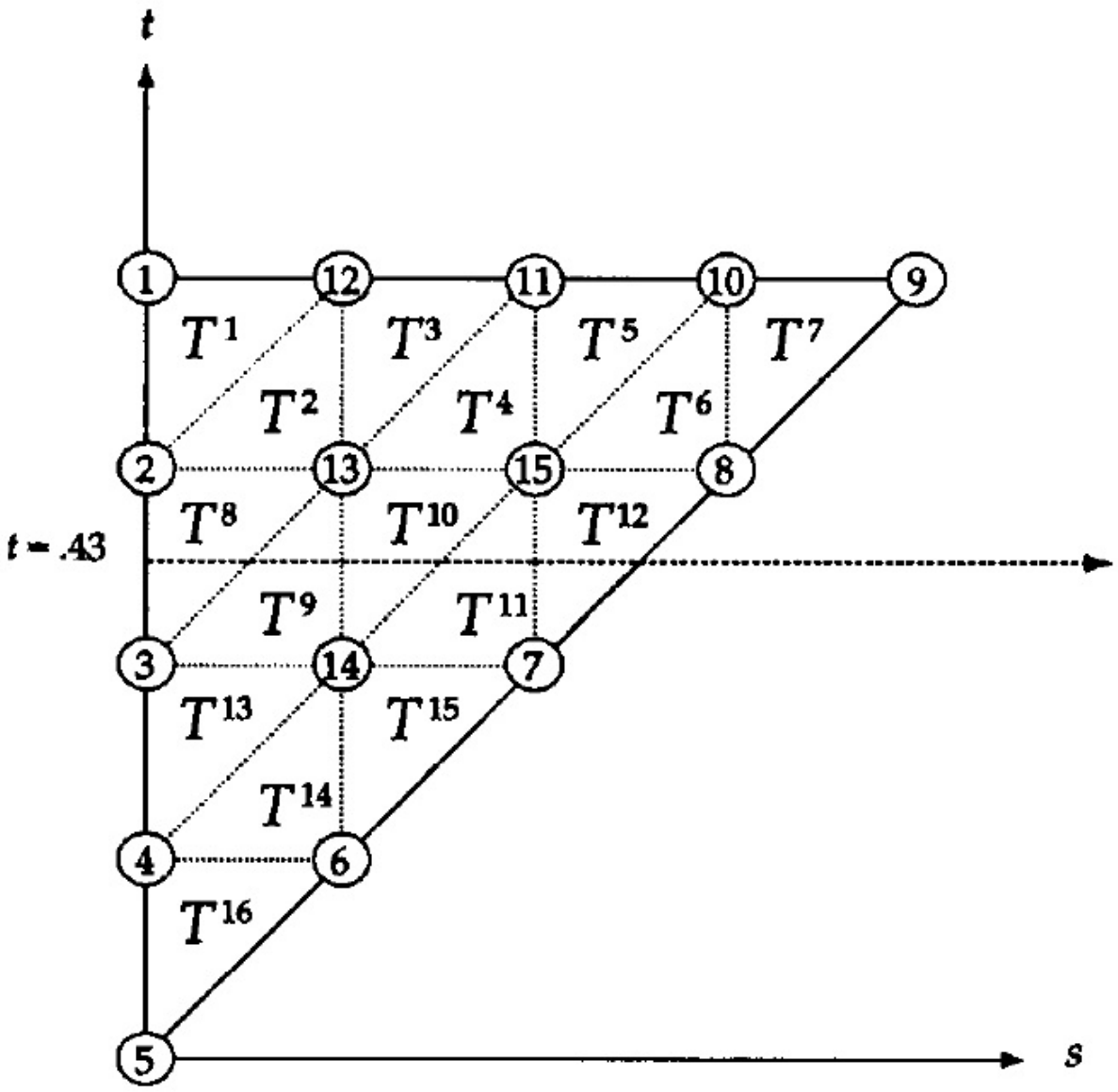


Autocorrelations are small beyond the lag of 300 msec.

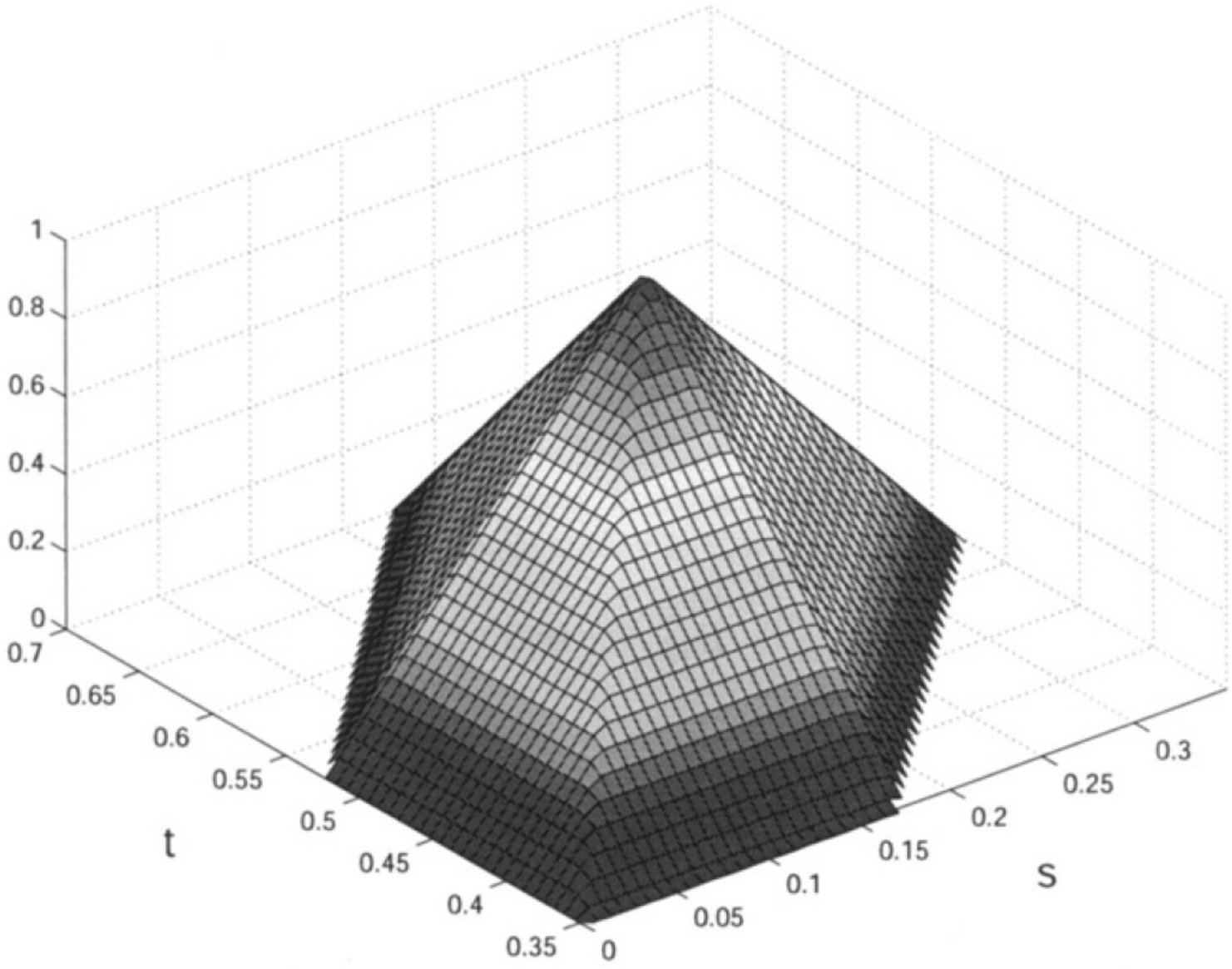
The finite element basis for $\beta(s, t)$

- The finite element basis for functions defined over two arguments uses piecewise linear functions defined over a triangular mesh.
- Triangular meshes can easily adapt to complex boundaries. Our problem here is particularly easy.
- The coefficient matrices become more and more sparse as the number of triangles increases.
- Sparse matrix computation makes for fast solutions.

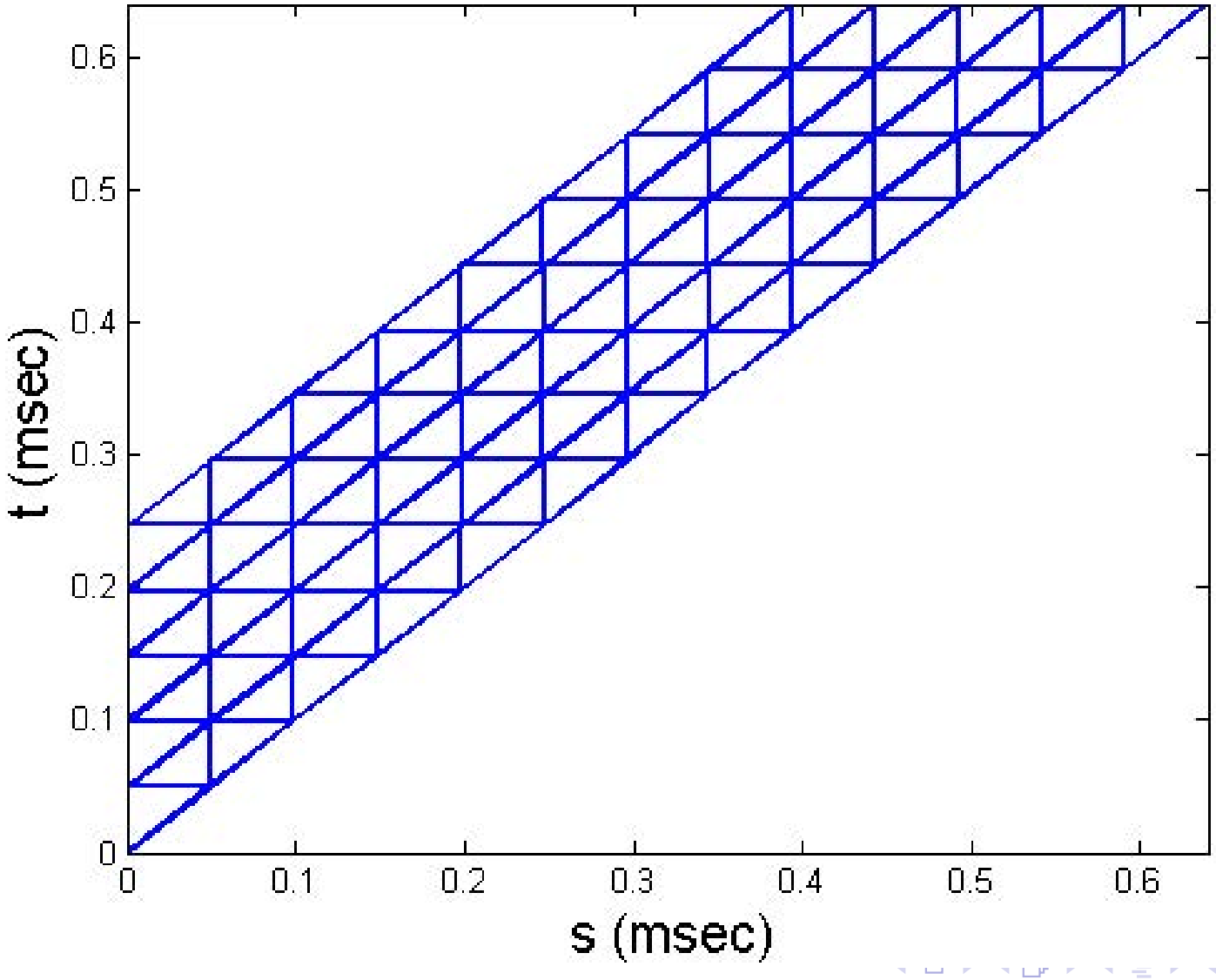
Complete triangular mesh for lip data



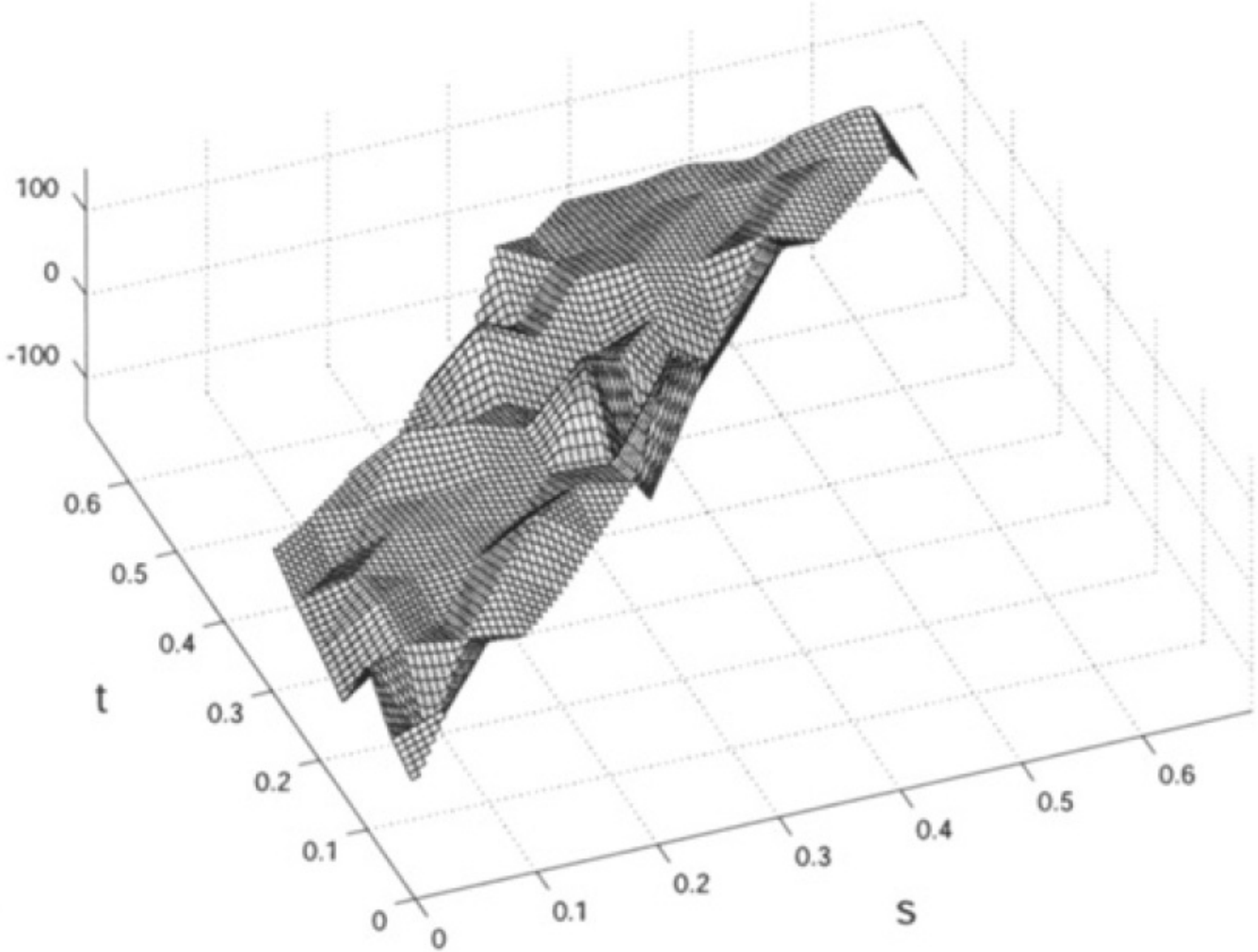
The basis function at node 13



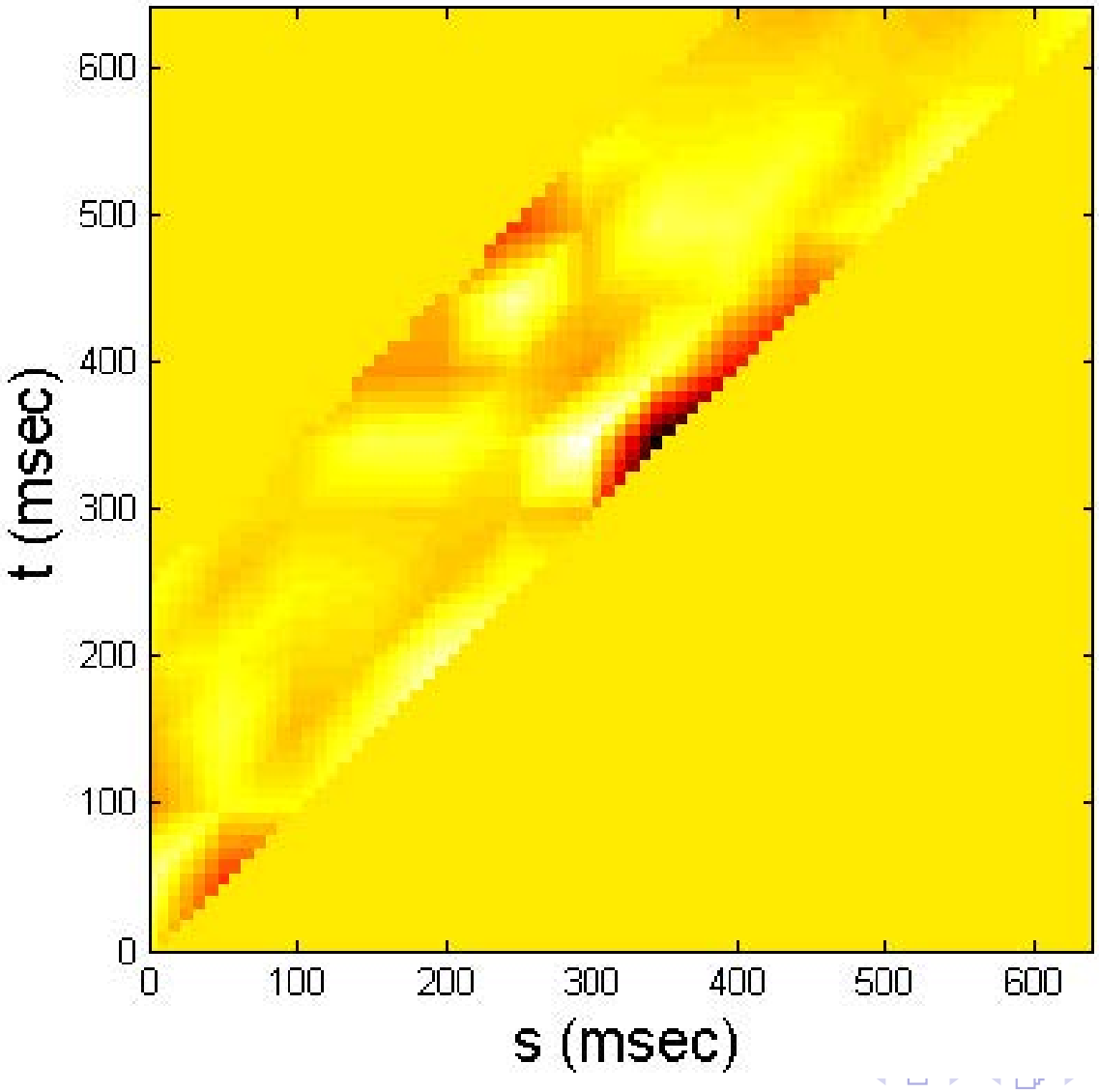
Triangular mesh for lip data



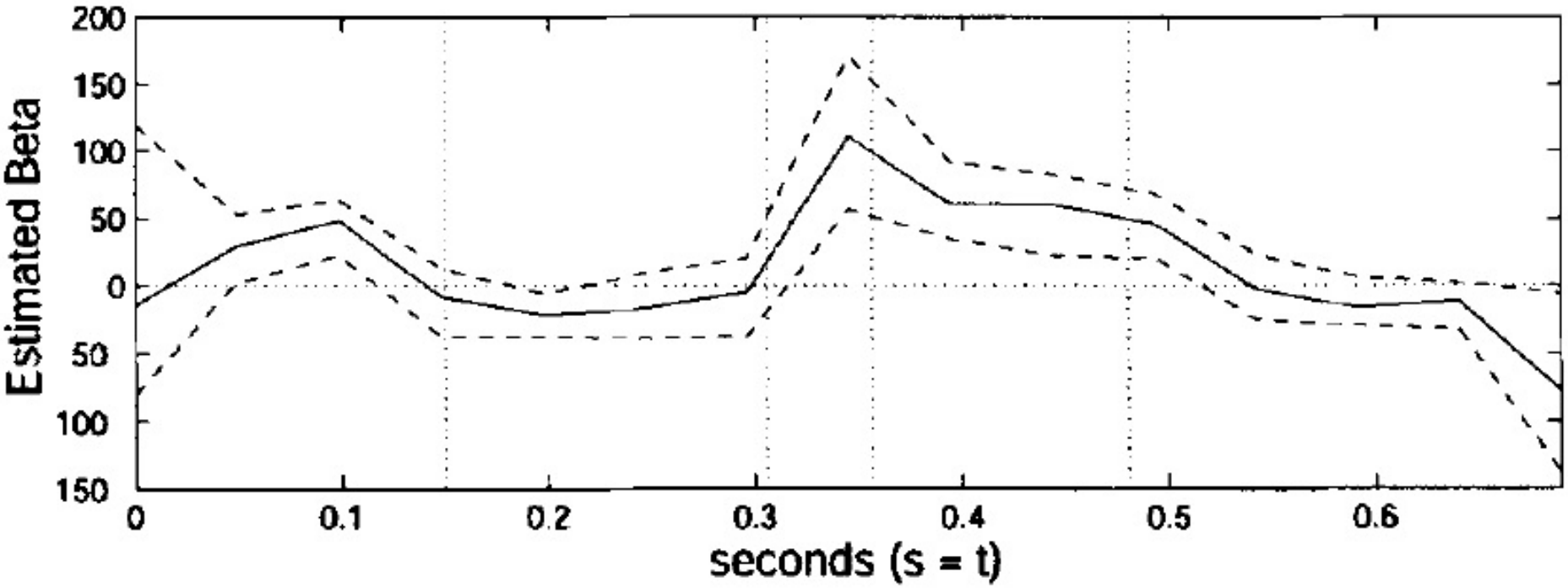
Lip data: $\hat{\beta}(s,t)$ with $\delta = 300msec$



Lip data: $\hat{\beta}(s,t)$ with $\delta = 250msec$



Lip data: $\hat{\beta}(t,t)$ with with 95% bootstrap CIs



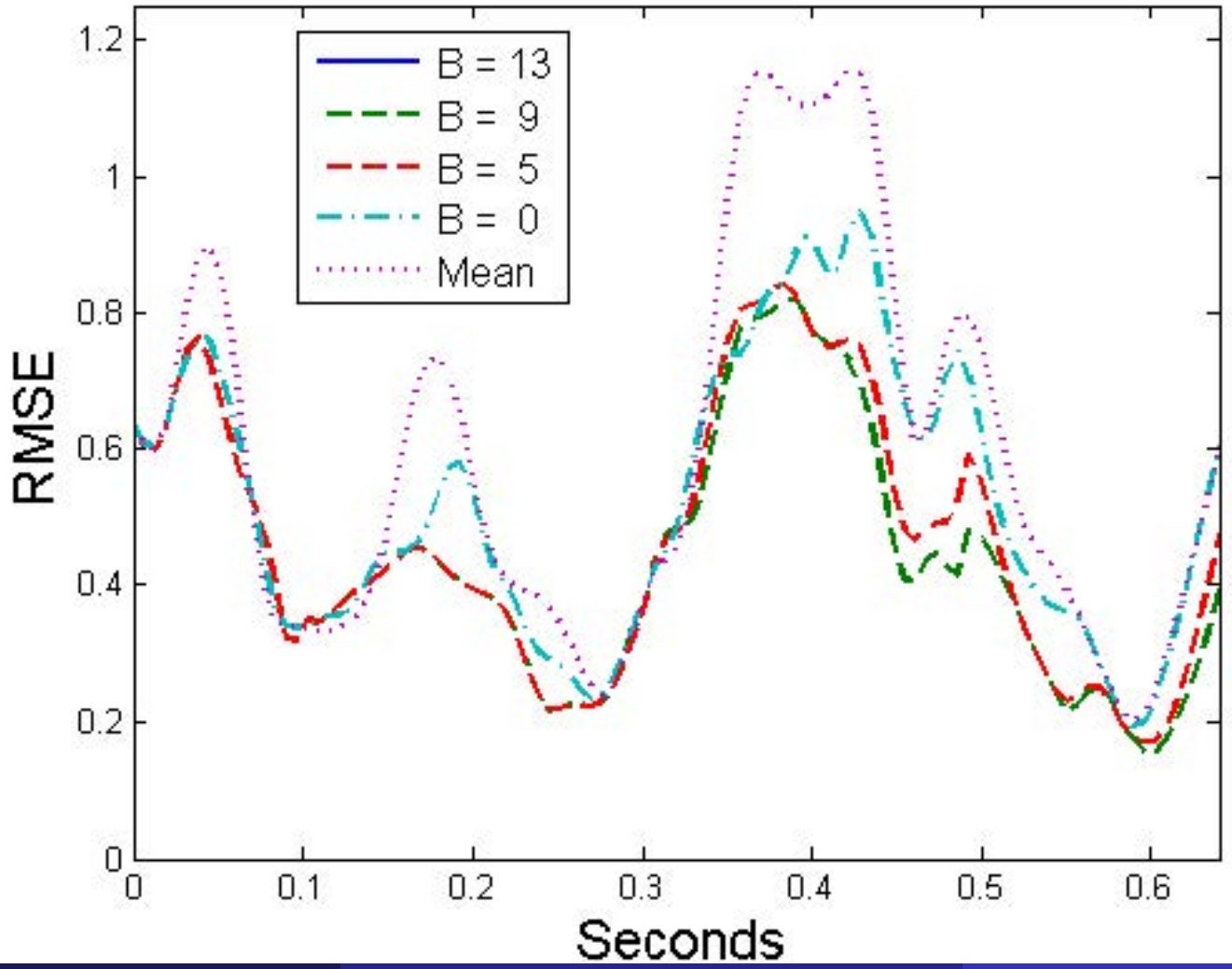
What do we see?

- As expected, the muscle activation is the most influential around the lip closure times: at both ends and between 300 and 400 msec.
- There is a ridge of influence along the diagonal, whose width indicates a transduction delay of about 50 msec.
- There are peaks deeper in the triangular domain: covariation of EMP events and lip displacements separated by longer delays.

An example peak

- Peak around $s = 250$ msec and $t = 500$ msec:
  - The agonist depressing activity of the DLI ends at around 250 msec, and this determines the position reached by the lower lip for the production of the /o/.
  - The amplitude of the upward movement that follows depends on this position.
  - And so does the deceleration during the second part of this movement that is necessary to allow for an adequate closing of the lips for the second /b/.
- A good illustration of how the pronunciation of a phoneme is determined by the chain of phonemes within which it is embedded.

Lip data: Std. err. of estimates for various lags



FLMs with functional responses

- When both response and covariates are functional, there are a lot of modeling possibilities.
- Using the full variation in a functional covariate $z(s)$ is only useful if we regularize the solution, either at the level of the regression coefficient $\beta(s,t)$, or at the level of the response $y(t)$.
- Using the full variation in the covariate usually only makes sense for periodic problems.
- A feed-forward model is more likely for non-periodic functional regressions.
- Estimating the amount of functional history to use is an important issue.
- There is a lot more work to do in this exciting area!

Where do we go from here?

- What if we want to use derivatives, $Dy(t)$ and $Dz(t)$, in our models?
- What about functional variables that interact with each other, such as what we find in input/output systems with feedback?
- Nonlinear functional models?
- We have only tickled the ear of the elephant.